

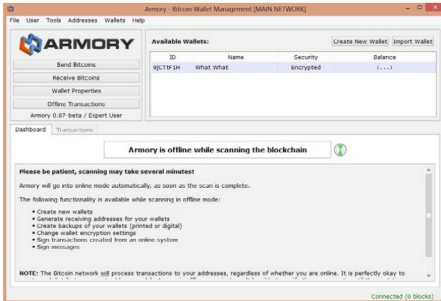


Figure 1: Armory is a full featured Bitcoin client that provides many advanced features

clients also available, these are generally not considered trustworthy or secure despite all security measures, as your Bitcoins are hosted on someone else's server. Modern day mobile devices now offer innovative ways of using Bitcoin to initiate a transaction between two devices with methods such as a QR code or an NFC tap.

Basically, when you install a Bitcoin client, a wallet is created for you. This wallet is associated with your Bitcoin address which looks like a long string of numbers and letters, for example:

1PC9aZC4hNX2nmrt7uHTFYAS3hRbph4UN

This address is used to send or receive Bitcoins from another person. Each wallet can be associated with as many addresses as possible. Actually, you should use each address for only one transaction. When this wallet is created, the Bitcoin client will ask you to encrypt it and create an associated password and a private key for accessing the wallet. This private key will allow you to spend Bitcoins from your wallet but anyone can monitor the balance in your wallet.

How it works

One might wonder what it is that gives value to a Bitcoin as opposed to normal currency. Why can it not be simply copied and spent multiple times? Whenever a Bitcoin transaction is intended between two parties, the transaction is broadcast to everyone on the network by the person who is responsible for sending the money to the recipient. This transaction is then verified and validated over the next couple of hours. This process thus avoids double

spending, as each verified transaction is then processed and included in the block-chain by the miners (the people who process all this information in return for transaction fees that are allocated to a miner, which may or may not be included).

The block-chain thus acts like a public ledger of all the Bitcoins that have been spent over time and keeps increasing in size.

Bitcoin clients

The table below gives some popular Bitcoin clients available on various platforms. Some of these are quite heavy due to the large size of the block-chain, while others like Electrum, Mycelium and online clients are relatively easy to use.

Bitcoin-Qt, for example, is the full featured thick client which downloads the whole block-chain using the official Bitcoin code. It requires significant network bandwidth and is quite heavy on system resources. Other third party clients, which keep the private keys on your local machine but use the block-chain that is hosted on a server, provide a relatively manageable compromise between security and speed. While the full featured clients are the most secure, it is not possible for everyone to have the patience and speed to run them on their machines. Each of these clients provides a simple set of instructions for installation on your machine and some of them are quite easy to use, even for those who are non-technical. Installation is similar to any other software. However, detailed instructions can be

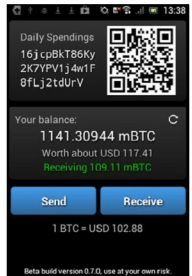


Figure 2: The Mycelium Bitcoin client provides quick Bitcoin transactions appropriate for a mobile client

Full featured desktop software clients	Mobile clients (Android / iOS)	Online Web wallets	Other clients (command line and GUI-less interfaces)
Bitcoin-Qt MultiBit Armory Electrum	Bitcoin Wallet Electrum Mycelium Wallet Blockchain.info	Coinbase BIPS	Subvertx bitcoind Gocoin